Mohammed Eid Abdelmeguid

AI Engineer | Machine Learning Engineer

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Professional Summary

AI/ML engineer with experience in Python backend development, retrieval-augmented generation (RAG), large language model (LLM) applications, and AI agent workflows. Contributed proxy-mode and document-ingestion features during an AI internship at Sprints. Built multi-agent applications and Arabic NLP systems; co-authored an ArabicNLP 2026 paper reporting blind-test results for spoken visual question answering.

Technical Skills

Programming and Data: Python, SQL, Pandas, NumPy, data cleaning, preprocessing, PostgreSQL
Generative AI and Retrieval: RAG, LangGraph, AI agents, document ingestion, chunking, embeddings, semantic search, pgvector, Qdrant, structured LLM outputs
Machine Learning and Evaluation: Scikit-learn, XGBoost, TensorFlow/Keras, feature engineering, PCA, grid search, classification, retrieval evaluation (Recall@K, NDCG@K, MAP); familiarity with PyTorch
NLP and Multimodal AI: Arabic NLP, CAMeL Tools, named entity recognition, BERTopic, knowledge graphs (NetworkX), Whisper speech recognition, vision-language models
Backend Engineering: FastAPI, Django, Flask, REST APIs, Celery, Redis, SQLAlchemy/SQLModel, unit testing
Deployment and Tools: Docker, AWS EC2, Vercel, Git, GitHub Actions, Linux, Ollama, Groq API

Professional Experience

AI Intern | Sprints
Oct 4, 2026 - Nov 15, 2026
SprintFlow Digital Twin - Proxy Mode and RAG Data Ingestion

- Implemented document ingestion from file upload through parsing, chunking, embedding, and indexing for PDF, DOCX, plain-text, and Markdown files.
- Built access-aware document retrieval with source citations using FastAPI, LangGraph, PostgreSQL, and Qdrant.
- Developed owner-controlled proxy replies and the knowledge used to ground them, with conversation-thread continuity and scoped access to private context.

Full Stack Developer Intern | NfrtiX
Mar 2025 - Apr 2025

- Built an internal Django and JavaScript application to support operational workflows.

Freelance Backend Developer | StockGo, Qatar
Nov 2024 - Apr 2025

- Built and deployed a Django REST API and PostgreSQL-backed analytics dashboard for a live inventory-tracking product.

Backend Developer Intern | Venture Appital
Oct 2024 - Feb 2025

- Developed modular Flask REST microservices and unit tests.

Projects

AI Compass | Multi-Agent RAG and Recommendation Platform
May 2026 - Present

- Led a team of five building an AI news platform with seven ingestion sources, five enrichment agents, semantic search, and a streaming RAG assistant.
- Implemented pgvector candidate retrieval, weighted ranking, and MMR diversification with an offline NDCG@10/MAP evaluation harness; integrated Django, Celery/Redis, and Groq-served LLMs.
- Deployed the frontend on Vercel and backend on AWS EC2 with PostgreSQL.

AynVQA-MSA | Arabic Spoken Visual Question Answering
Jul 2026 - Oct 2026

- Implemented a modular Whisper speech-recognition, schema-constrained parsing, conditional repair, and joint vision-language answer-selection pipeline.
- Co-authored published results: **65.6% accuracy on 1,000 blind-test examples**, versus 79.0% on 500 development examples; analyzed the recording-condition shift and a caching reproducibility issue.

Palestinian Cultural Knowledge Platform | Arabic NLP and Retrieval
Jun 2026 - Aug 2026

- Owned ingestion, Arabic NLP, retrieval, and knowledge-graph work on a two-person project; built quality scoring, normalization, and persistent MinHash/LSH deduplication.
- Published 882 Arabic/English documents (about 890K words) on Hugging Face with gated file access; built a graph of 13,063 canonicalized entities and a retriever reporting Recall@5 of 0.93 over 1,282 passages.

Fitness Activity Tracker | Sensor-Based Machine Learning
Mar 2026 - Apr 2026

- Preprocessed wearable sensor data, engineered PCA/FFT features, compared five classifiers across five feature sets, and implemented independent repetition counting on a three-person team.

Publications

Digilians at ImageEval 2026 Shared Tasks: Modular Speech Parsing and Joint Multimodal Reasoning for Arabic Spoken VQA
Ahmed Eid Hassan, Nour Eldeen Hossam, Ahmed Hossam Rashad, Abdelrhman Mahmoud Fawzy, **Mohammed Eid Abdelmeguid**, and Mohamed Hamdy Mohamed. *ArabicNLP 2026: Shared Tasks*, pp. 766--770, Oct 2026. DOI: 10.18653/2026.arabicnlp-sharedtasks.96.

Education

AI and Data Science Diploma | Digilians
MCIT / Egyptian Military Academy scholarship | Dec 2025 - Sep 2026

BS in Communication and Electronics Engineering | Alexandria University
Faculty of Engineering | Sep 2019 - Jun 2024 | GPA: 3.13/4.0
ADAS graduation project: A+, ranked 5th of about 60 in the Valeo competition. Led software architecture and computer vision; mentored by Valeo and sponsored by ITIDA's Egypt Makes Electronics program.

Certifications

AWS Certified Machine Learning Engineer - Associate | Amazon Web Services | Sep 2026
Huawei Certified ICT Associate - AI (HCIA-AI) | Huawei | Aug 2026
IBM AI Engineering Professional Certificate | IBM / Coursera | Jun 2026
Machine Learning Specialization | Stanford / DeepLearning.AI | May 2026
Generative AI for Data Science with Copilot | Microsoft / Coursera | Apr 2026
Microsoft Data Engineer (159 hours) | DEPI | Oct 2024